

SOME PROPERTIES OF SOMERVILLE'S MULTIPLE RANGE SUBSET SELECTION PROCEDURE FOR THREE POPULATIONS

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Summary: Somerville (1984) has recently proposed a
 multiple range subset selection procedure
 for the case when sample means of size n
 are obtained from k normal populations with
 common known variance. It is shown here that
 when $k=3$ the probability of correct selection
 of this procedure is between that of Gupta's
 procedure and P^* and strong evidence is
 presented that the expected subset size of the
 procedure is lower than that of Gupta's pro-
 cedure uniformly in parameter values for all
 $P^* \leq 0,98$.

1. INTRODUCTION

Suppose we have independent samples, each of size n , from k popu-
lations. Call the population with the largest mean the "best".

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Using the sample data we want to select a subset of populations which includes the best one with a prescribed probability P^* .

A procedure for doing this was proposed by Gupta (1956, 1965). In this paper we consider only the case where the i -th population is $N(\mu_i, \sigma^2)$ -distributed, $i=1,2,\dots,k$. We suppose σ^2 known and $\underline{\mu} = (\mu_1, \dots, \mu_k)$ unknown. In this case Gupta's procedure selects all those populations whose sample means are within distance $d\sigma/n^{1/2}$ from the maximal sample mean, where d depends on k and P^* and satisfies

$$(1) \quad \int_{-\infty}^{\infty} \Phi(x+d)^{k-1} \phi(x) dx = P^*$$

with ϕ and Φ the $N(0,1)$ -density and distribution functions respectively.

Somerville (1984) proposed the following procedure: Let $d_0=0$ and let $d_1, d_2, \dots$ be an increasing sequence of numbers depending on P^* (but not on k) to be specified further below. Then, if the range of all k sample means is less than $d_{k-1}\sigma/n^{1/2}$, include all k populations in the selected subset. Else, eliminate the population with the smallest sample mean and calculate the range of the remaining sample means. If this is less than $d_{k-2}\sigma/n^{1/2}$ include the populations corresponding to the $k-1$ largest sample means. Else, eliminate the population with the second smallest sample mean, and so forth. This procedure is called the multiple range subset selection procedure.

Somerville (1984) gives tables for $d_1, d_2, \dots$ and reports numerical and Monte Carlo results which show that his procedure compares favourably with Gupta's in terms of expected subset sizes. It is the aim of this paper to establish some properties of this procedure for the case of three populations. Specifically, in §3 we show that for any $\underline{\mu}$ -value, the probability of correct selection (PCS) of the multiple range procedure is between P^* and the PCS of Gupta's procedure. In §4 we present evidence that there exists some number Q^* such that if $P^* \leq Q^*$ then the expected subset size (ESS) of the multiple range procedure is

less than or equal to the ESS of Gupta's procedure uniformly in $\underline{\mu}$. Numerical work shows that, Q^* is approximately 0.98. Moreover, for $P^* > Q^*$, the ESS of the multiple range procedure is actually larger than that of Gupta's procedure for only small parameter regions and only by a small amount.

2. GEOMETRICAL REPRESENTATIONS

Henceforth we restrict attention to the case of $k=3$ normal populations with the same variance σ^2 . We assume that $\sigma/\sqrt{n}=1$ without loss of generality. Let X_1, X_2, X_3 denote the sample means of the three populations. Then X_i is $N(\mu_i, 1)$ -distributed, $i=1, 2, 3$ and they are mutually independent. Introduce three new variables Y_1, Y_2, Y_3 by

$$\begin{aligned} Y_1 &= (X_1 - X_2 + X_3)/\sqrt{2} \\ (2) \quad Y_2 &= (-2X_1 + X_2 + X_3)/\sqrt{6} \\ Y_3 &= (X_1 + X_2 + X_3)/\sqrt{3}. \end{aligned}$$

Then Y_i is $N(\nu_i, 1)$ -distributed, $i=1, 2, 3$ and they are also mutually independent. ν_1, ν_2, ν_3 are related to μ_1, μ_2, μ_3 in the obvious way.

We first look at Gupta's procedure and see what it becomes in terms of the Y_i 's. The inverse transform of (2) is

$$\begin{aligned} X_1 &= Y_1/\sqrt{2} - Y_2/\sqrt{6} + Y_3/\sqrt{3} \\ (3) \quad X_2 &= -Y_1/\sqrt{2} + Y_2/\sqrt{6} + Y_3/\sqrt{3} \\ X_3 &= Y_1/\sqrt{2} + Y_2/\sqrt{6} + Y_3/\sqrt{3}. \end{aligned}$$

The region $X_1 > X_2 > X_3$ translates to $Y_2 < Y_1/\sqrt{3} < 0$. In this region Gupta's procedure selects all three populations iff $X_1 - X_3 < d$ which translates to $Y_2 > (-\sqrt{2}d - Y_1)/\sqrt{3}$. If this is not the case but $X_1 - X_2 < d$, then populations 1 and 2 are selected - which translates to $Y_2 > (Y_1 - \sqrt{2}d)/\sqrt{3}$. Finally, if this does not happen

then only population 1 is selected. In similar fashion one can derive the subsets in Y -space corresponding to various other orderings of the X_i 's and subsets selected by Gupta's procedure. Notice that Y_3 drops out entirely and Figure 1 shows the resulting subsets in (Y_1, Y_2) -space.

Turning to the multiple range procedure if $X_1 > X_2 > X_3$ then all three populations are selected if $X_1 - X_3 < d_2$ which corresponds to $Y_2 > (\sqrt{2}d_2 - Y_1)/\sqrt{3}$. If this does not happen but $X_1 - X_2 < d_1$ then populations 1 and 2 are selected and this translates to $Y_2 > (Y_1 - \sqrt{2}d_1)/\sqrt{3}$. Otherwise only population 1 is selected. Arguing similarly for other cases, one sees that the multiple range procedure can be represented in (Y_1, Y_2) -space as in Figure 2.

Figures 1 and 2 show the situation for $P^* = 0,8$ in which case $d = 1,65241$, $d_1 = 1,19023$ and $d_2 = 1,84492$. In general d is chosen to make the PCS of Gupta's procedure equal to P^* when $\nu_1 = \nu_2 = 0$. For the multiple range procedure we take $d_1 = \sqrt{2} \Phi^{-1}(P^*)$ if $P^* > 0,5$ and $d_1 = 0$ otherwise. Then d_2 is chosen to make the PCS of this procedure also equal to P^* when $\nu_1 = \nu_2 = 0$. This leads to $d_1 < d < d_2$ for $P^* < 1$. When $P^* \leq 0,5$ the regions where two populations are selected disappear entirely; when P^* decreases to $\frac{1}{3}$, this also happens for the region where all three populations are selected. We discuss the PCS further in §3 below. Tables of values of d , d_1 and d_2 at other P^* values can be found in Somerville (1984).

It is possible to discuss a family of procedures of Somerville's type in which d_1 varies between $\max(0, \sqrt{2} \Phi^{-1}(P^*))$ and d , and d_2 is chosen in the same way as above; then Gupta's procedure is the extreme member corresponding to $d_1 = d = d_2$. Many of the results below are valid for this larger class of procedures. However, to keep matters simple we discuss only the case $d_1 = \max(0, \sqrt{2} \Phi^{-1}(P^*))$.

It is easy to see that $\mu_1 > \mu_2 \geq \mu_3$ corresponds to $\nu_2 < \nu_1/\sqrt{3} \leq 0$ and $\mu_1 > \mu_3 \geq \mu_2$ to $\nu_2 < -\nu_1/\sqrt{3} \leq 0$. Thus, population 1 being best corresponds to $\nu_2 < -|\nu_1|/\sqrt{3}$ which is the 120° sector symmetric

around the vertical axis in the lower half plane. The 60° sector to the left of the vertical axis corresponds to population 1 being the best and population 2 being second best and the 60° sector to the right of the vertical axis corresponds to population 1 being best and 3 being second best. The outer boundaries of these sectors are $v_2 = v_1/\sqrt{3}$ and $v_2 = -v_1/\sqrt{3}$ which correspond to $\mu_1 = \mu_2 > \mu_3$ and $\mu_1 = \mu_3 > \mu_2$ respectively (i.e. the two largest means equal). The lower half of the vertical axis corresponds to $\mu_1 > \mu_2 = \mu_3$ (i.e. the two smallest means equal). This half line will also be called axis 1. The bisectors between axis 1 and the boundaries of the sector in which population 1 is the best correspond to the case of equi-spaced means.

In the same way axes 2 and 3 (rotated 120° backwards and forwards respectively from axis 1) are central to sectors in which populations 2 and 3 are best respectively and these sectors can be divided into subsectors with analogous meanings. It may also be noted that axes 1, 2 and 3 are the projections of the μ_1 , μ_2 and μ_3 axes of the original co-ordinate system onto the plane $v_3 = 0$. The origin $v_1 = v_2 = 0$ corresponds to $\mu_1 = \mu_2 = \mu_3$. Figure 3 summarizes these facts.

From Figures 1 and 2 we see that the procedures are invariant with respect to reflection in axes 1, 2 and 3 and also with respect to 120° rotations. Together with the symmetry of the circular bivariate normal distribution this implies that the PCS and the ESS of the procedures are the same at (v_1, v_2) points which are reflections of each other in the axes 1, 2 and 3 or 120° degree rotations from each other. Thus we need only determine them in one 60° sector; in the rest of the paper we will take (v_1, v_2) in the sector $v_1 \geq 0$, $v_2 \leq -v_1/\sqrt{3}$ (which is half of the sector where population 1 is best).

3. PROBABILITY OF CORRECT SELECTION

Theorem 1:

The PCS of the multiple range procedure is at least P^* for any $\underline{\mu}$.

Proof:

As explained above, we consider PCS a function of (ν_1, ν_2) and we need only treat the case $\nu_2 \leq -\nu_1/\sqrt{3}$, $\nu_1 \geq 0$. Then population 1 is the best and it is included in the selected subset iff (Y_1, Y_2) is below the polygon ABCDEFG in Figure 4. We need to show that the probability content of this region with respect to a unit circular bivariate normal distribution centred at (ν_1, ν_2) is at least P^* . If we increase ν_2 holding ν_1 fixed, this probability evidently decreases. Hence it suffices to take $\nu_2 = -\nu_1/\sqrt{3}$.

Consider first the case $P^* \geq \frac{1}{2}$. By the choice of d_1 the probability content below the line HG in Figure 4 is P^* for any ν_1 . We need therefore only show that the probability content of the region a is at least that of b in Figure 4, i.e. that

$$\int_a \phi(y_1 - \nu_1) \phi(y_2 + \nu_1/\sqrt{3}) dy_1 dy_2 \geq \int_b \phi(y_1 - \nu_1) \phi(y_2 + \nu_1/\sqrt{3}) dy_1 dy_2$$

or

$$(4) \quad \int_a \phi(y_1) \phi(y_2) \exp[\nu_1(y_1 - y_2/\sqrt{3})] dy_1 dy_2 \\ \geq \int_b \phi(y_1) \phi(y_2) \exp[\nu_1(y_1 - y_2/\sqrt{3})] dy_1 dy_2.$$

By Jensen's inequality the left-hand side is at least

$$\exp[\nu_1(m_1 - m_2/\sqrt{3})] \int_a \phi(y_1) \phi(y_2) dy_1 dy_2$$

while the right-hand side is at most

$$\exp[\nu_1(s_1 - s_2/\sqrt{3})] \int_b \phi(y_1) \phi(y_2) dy_1 dy_2$$

where

$$m_i = \int_a y_i \phi(y_1) \phi(y_2) dy_1 dy_2 / \int_a \phi(y_1) \phi(y_2) dy_1 dy_2$$

for $i=1,2$ and (s_1, s_2) is the point in b at which $\exp[\nu_1(y_1 - y_2/\sqrt{3})]$ is maximized. Since $\int_a \phi(y_1)\phi(y_2)dy_1dy_2 = \int_b \phi(y_1)\phi(y_2)dy_1dy_2$ we need only show that $\exp[\nu_1(m_1 - m_2/\sqrt{3})] \geq \exp[\nu_1(s_1 - s_2/\sqrt{3})]$. The contours of $\exp[\nu_1(y_1 - y_2/\sqrt{3})]$ are lines perpendicular to HG and by symmetry (m_1, m_2) is on the contour through the origin. (s_1, s_2) is either the point H or the point B; in both cases it is on a lower contour and the theorem follows.

For the case $\frac{1}{3} \leq P^* < \frac{1}{2}$ replace the line HG in the above argument by a line drawn below and parallel to FG such that there is P^* probability below it. With minor changes the argument then remains valid.

Theorem 2:

The PCS of the multiple range procedure is at most that of Gupta's procedure.

Proof:

Referring to Figure 5 we see that the difference between the two procedures with respect to inclusion of population 1 is that Gupta's includes it when (Y_1, Y_2) is in one of the sets c_1 or c_2 while the multiple range includes it when (Y_1, Y_2) is in one of the sets a_1 or a_2 . We need to show that

$$\int_{c_1 \cup c_2} \phi(y_1 - \nu_1)\phi(y_2 - \nu_2)dy_1dy_2 \geq \int_{a_1 \cup a_2} \phi(y_1 - \nu_1)\phi(y_2 - \nu_2)dy_1dy_2$$

where $\nu_2 \leq -\nu_1/\sqrt{3}$ and $\nu_1 \geq 0$. By symmetry we have

$$\int_{c_2} \phi(y_1 - \nu_1)\phi(y_2 - \nu_2)dy_1dy_2 = \int_{c_1} \phi(y_1 + \nu_1)\phi(y_2 - \nu_2)dy_1dy_2.$$

A similar relation holds with c_1 and c_2 replaced by a_1 and a_2 . Therefore, it suffices to show that

$$\int_{c_1} \phi(y_2 - \nu_2) [\phi(y_1 - \nu_1) + \phi(y_1 + \nu_1)] dy_1 dy_2 \\ \geq \int_{a_1} \phi(y_2 - \nu_2) [\phi(y_1 - \nu_1) + \phi(y_1 + \nu_1)] dy_1 dy_2 .$$

This simplifies to

$$\int_{c_1} \phi(y_1) \phi(y_2) g(y_1, y_2) dy_1 dy_2 \geq \int_{a_1} \phi(y_1) \phi(y_2) g(y_1, y_2) dy_1 dy_2$$

where $g(y_1, y_2) = \exp(\nu_2 y_2) \cosh(\nu_1 y_1)$ which is minimized over c_1 and maximized over a_1 at the point A. Since

$$\int_{c_1} \phi(y_1) \phi(y_2) dy_1 dy_2 = \int_{a_1} \phi(y_1) \phi(y_2) dy_1 dy_2 \quad \text{the theorem follows.}$$

Figure 6 shows the contours of the difference between the PCS's of Gupta's procedure and the multiple range for $P^* = 0,8$. The contours for other P^* -values are quite similar. It is clear that the maximal difference occurs on the lines where the largest two means are equal. In the limit as one moves out on these lines the difference tends to $\Phi(d/\sqrt{2}) - \Phi(d_1/\sqrt{2})$.

4. EXPECTED SUBSET SIZE

Let G and R denote the expected subset sizes of Gupta's procedure and the multiple range procedure respectively. Of course these are functions of (ν_1, ν_2) . Useful expressions for G , R and $D = G - R$ will now be developed. We have

$$(5) \quad G = 1 + T_{2G} + 2T_{3G}$$

where T_{2G} and T_{3G} are the probabilities of 2 and 3 populations respectively being selected by Gupta's procedure. Then

$$T_{2G} = \Sigma \int_{c_i} \phi(y_1 - \nu_1) \phi(y_2 - \nu_2) dy_1 dy_2$$

$$T_{3G} = \Sigma \int_{a_i} \phi(y_1 - \nu_1) \phi(y_2 - \nu_2) dy_1 dy_2$$

with the regions c_i and a_i as defined in Figure 7a.

Let us denote the point (ν_1, ν_2) by P_1 and let P_2 be its reflection in axis 1, i.e. $P_2 = (-\nu_1, \nu_2)$. Also, let P_3, P_4 and P_5, P_6 be points whose position with respect to axes 2 and 3 are the same as those of P_1, P_2 with respect to axis 1. So

$$(6) \quad P_3 = ((-\nu_1 + \sqrt{3}\nu_2)/2, (-\sqrt{3}\nu_1 - \nu_2)/2) \quad P_4 = ((\nu_1 + \sqrt{3}\nu_2)/2, (\sqrt{3}\nu_1 - \nu_2)/2)$$

$$P_5 = ((-\nu_1 - \sqrt{3}\nu_2)/2, (\sqrt{3}\nu_1 - \nu_2)/2) \quad P_6 = ((\nu_1 - \sqrt{3}\nu_2)/2, (-\sqrt{3}\nu_1 - \nu_2)/2).$$

We abbreviate $\phi(y_1 - \alpha_1)\phi(y_2 - \alpha_2)$ with $(\alpha_1, \alpha_2) = P_i$ by f_i , $i = 1, 2, \dots, 6$. Then, by symmetry, we see that

$$\int_{a_2} f_1 dy_1 dy_2 = \int_{a_1} f_2 dy_1 dy_2$$

$$\int_{a_3} f_1 dy_1 dy_2 = \int_{a_1} f_5 dy_1 dy_2$$

$$\int_{a_4} f_1 dy_1 dy_2 = \int_{a_1} f_4 dy_1 dy_2$$

$$\int_{a_5} f_1 dy_1 dy_2 = \int_{a_1} f_3 dy_1 dy_2$$

$$\int_{a_6} f_1 dy_1 dy_2 = \int_{a_1} f_6 dy_1 dy_2.$$

Hence

$$(7) \quad T_{3G} = \int_{a_1} f dy_1 dy_2$$

where

$$(8) \quad f = \Sigma f_i.$$

In the same way one shows that

$$(9) \quad T_{3G} = \int_{a_i} f \, dy_1 dy_2 \quad \text{and} \quad T_{2G} = \int_{c_i} f \, dy_1 dy_2$$

for any i . We find it somewhat convenient to work with a_1 and a_4 and c_1 . Hence we have

$$(10) \quad G = 1 + \int_{c_1 \cup a_1 \cup a_4} f \, dy_1 dy_2.$$

In the same way we obtain

$$(11) \quad R = 1 + \int_{c'_1 \cup a'_1 \cup a'_4} f \, dy_1 dy_2$$

with c'_1 , a'_1 and a'_4 as defined in Figure 7b. The two procedures are shown together in Figure 7c and it is clear that

$$(12) \quad D = \int_c f \, dy_1 dy_2 - \int_b f \, dy_1 dy_2$$

with c and b as defined there.

Contours of D were computed from (12) and Figures 8 and 9 show these for $P^* = 0,8$ and $0,99$ respectively. We see that D is everywhere non-negative for $P^* = 0,8$. Numerical work shows that this continues to be the case for P^* up to just above $0,98$. In Figure 9 small regions (shaded in Figure 9) appear where D is negative at $P^* = 0,99$; the maximal negative value is quite small ($0,003$). For P^* larger than $0,99$ these negative regions persist. This numerical work suggests that there exists some number Q^* between $0,98$ and $0,99$ such that $D(\nu_1, \nu_2) \geq 0$ for all (ν_1, ν_2) whenever $P^* \leq Q^*$. The following argument strengthens this conjecture considerably.

Substituting into (8) and rearranging terms we get

$$f(y_1, y_2) = \phi(y_1)\phi(y_2)g(y_1, y_2)\exp(-1/2(\nu_1^2 + \nu_2^2))$$

with

$$g(y_1, y_2) = 2[\exp(\nu_2 y_2) \cosh(\nu_1 y_1) + \exp((- \sqrt{3} \nu_1 - \nu_2) y_2 / 2) \cosh((\nu_1 - \sqrt{3} \nu_2) y_1 / 2) + \exp((\sqrt{3} \nu_1 - \nu_2) y_2 / 2) \cosh((\nu_1 + \sqrt{3} \nu_2) y_1 / 2)].$$

This function has the following properties. Since $g(-y_1, y_2) = g(y_1, y_2)$ we see that g is symmetric with respect to axis 1; moreover, if (y_1, y_2) moves perpendicular to axis 1 towards this axis, then $g(y_1, y_2)$ decreases, reaching a minimum on this axis. Actually g behaves in the same way with respect to axis 2 and 3 also: it is symmetric with respect to reflection in these axes and decreases when its argument moves perpendicularly towards these axes. Finally, $g(y_1, y_2)$ is also convex in (y_1, y_2) .

The next argument is somewhat simpler if we replace $\int_b f dy_1 dy_2$ by $\int_a f dy_1 dy_2$ in equation (12) with a as defined in Figure 7c. These two integrals are equal in view of the symmetry of $g(y_1, y_2)$ and also $\phi(y_1)\phi(y_2)$ with respect to axis 3.

Refer to Figure 10 now. We want to find a sufficient condition under which

$$(13) \quad \int_c \phi(y_1)\phi(y_2)g(y_1, y_2)dy_1 dy_2 \geq \int_a \phi(y_1)\phi(y_2)g(y_1, y_2)dy_1 dy_2$$

for all (ν_1, ν_2) . From Jensen's inequality the left-hand side is at least

$$(14) \quad g(m_1, m_2) \int_c \phi(y_1)\phi(y_2)dy_1 dy_2$$

with

$$(15) \quad m_i = \int_c y_i \phi(y_1)\phi(y_2)dy_1 dy_2 / \int_c \phi(y_1)\phi(y_2)dy_1 dy_2$$

for $i=1,2$. Also, the right-hand side of (13) is at most

$$(16) \quad g(s_1, s_2) \int_a \phi(y_1) \phi(y_2) dy_1 dy_2$$

with (s_1, s_2) the point in a at which $g(y_1, y_2)$ is maximized. Since the integrals in (14) and (16) are equal (use $D(0,0) = 0$), (13) will follow if we have

$$(17) \quad g(m_1, m_2) \geq g(s_1, s_2)$$

for all (ν_1, ν_2) . It follows from the properties of g that it achieves its maximum over a along the upper boundary (the line $y_2 = (\sqrt{2}d_2 - y_1)/\sqrt{3}$); by the convexity the maximum must be at one of the end points. When (ν_1, ν_2) is within the 30° sector to the right of axis 1, the maximum occurs at the point $(d_2/\sqrt{2}, d_2/\sqrt{6})$; otherwise it occurs at $(0, \sqrt{2}d_2/\sqrt{3})$.

Consider the first of these two cases, i.e. $(s_1, s_2) = (d_2/\sqrt{2}, d_2/\sqrt{6})$. Let (m'_1, m'_2) be the projection of (m_1, m_2) onto axis 3. If (m'_1, m'_2) is further out along this axis than (s_1, s_2) , then (17) will be true since $g(y_1, y_2)$ decreases when (y_1, y_2) moves from (m_1, m_2) to (m'_1, m'_2) and also when (y_1, y_2) moves along axis 3 from (m'_1, m'_2) to (s_1, s_2) . Let Δ_1 denote the distance between (m'_1, m'_2) and (s_1, s_2) ; then a sufficient condition for $D(\nu_1, \nu_2) \geq 0$ for all (ν_1, ν_2) within a 30° sector next to axis 1 (and by symmetry also for 30° sectors on both sides of axes 2 and 3), is that $\Delta_1 \geq 0$.

Consider next the second case, i.e. where $(s_1, s_2) = (0, \sqrt{2}d_2/\sqrt{3})$. In the same way we find that a sufficient condition for $D(\nu_1, \nu_2) \geq 0$ with (ν_1, ν_2) in the complement of the regions above is that $\Delta_2 \geq 0$ where Δ_2 is the distance between the projection of (m_1, m_2) onto the y_2 -axis and (s_1, s_2) , i.e. $\Delta_2 = m_2 - \sqrt{2}d_2/\sqrt{3}$.

Letting $\Delta = \min(\Delta_1, \Delta_2)$ the condition $\Delta \geq 0$ is sufficient for $D(\nu_1, \nu_2) \geq 0$ for all (ν_1, ν_2) . Note that Δ is a function of P^* only. While it is somewhat difficult to handle analytically, it is easy

enough to compute. This was done and it was found that Δ was slightly negative for P^* in a small neighbourhood of 0,5 (where Δ_1 was negative) and more strongly negative for P^* above 0,78 (where Δ_2 was negative). So, while leading to significant results, this argument was not quite satisfactory.

To get better results, the regions a and c may be split into subregions which can be compared separately. This can be done in many ways. We describe only one briefly. Refer to Figure 11 and let a_1 be the sub-region of a shown with the line labelled ℓ_1 at the level $(2d_2 - 0,8d_1)/\sqrt{6}$ while a_2 is the remainder of a . Cut c by the line ℓ_2 parallel to the boundary of a and chosen such that

$$\int_{c_1} \phi(y_1)\phi(y_2)dy_1dy_2 = \int_{a_1} \phi(y_1)\phi(y_2)dy_1dy_2 .$$

Let c_2 be the remainder of c . Then, if (m_1, m_2) is the mean point in c_1 and (n_1, n_2) the mean point in c_2 , a sufficient condition for $D(\nu_1, \nu_2) \geq 0$ for all (ν_1, ν_2) is that $\Delta^* \geq 0$ where

$\Delta^* = \min(\Delta_1, \Delta_2, \Delta_3, \Delta_4)$ with the Δ_i 's as shown in Figure 11. Figure 12 shows Δ^* as a function of P^* and it is clear that $\Delta^* \geq 0$ for all $P^* \leq Q^*$ with $Q^* \geq 0,82$. We mention that the level of the cutting line ℓ_1 was chosen empirically to balance the various Δ_i 's to get the best result here.

More complicated subdivisions of the regions a and c were also done and the best result lead to a Q^* value of 0,93 with this approach.

5. CONCLUSION

Somerville's multiple range subset selection procedure has been shown to compare very favourably with Gupta's procedure in the case of three normal populations with common known variance. Open questions are:

- Does this finding extend to the case of unknown common variance and to the case of more than three populations?
- Does there exist a procedure which has an ESS which is uniformly below that of Gupta's procedure for all P^* values. Results in this direction would have significant implications for the study of optimality of subset selection procedures (see e.g. Gupta, S.S. and Miescke, K.J. (1981)).

6. ACKNOWLEDGEMENT

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FIGURE 1. Gupta's procedure in (Y_1, Y_2) -space

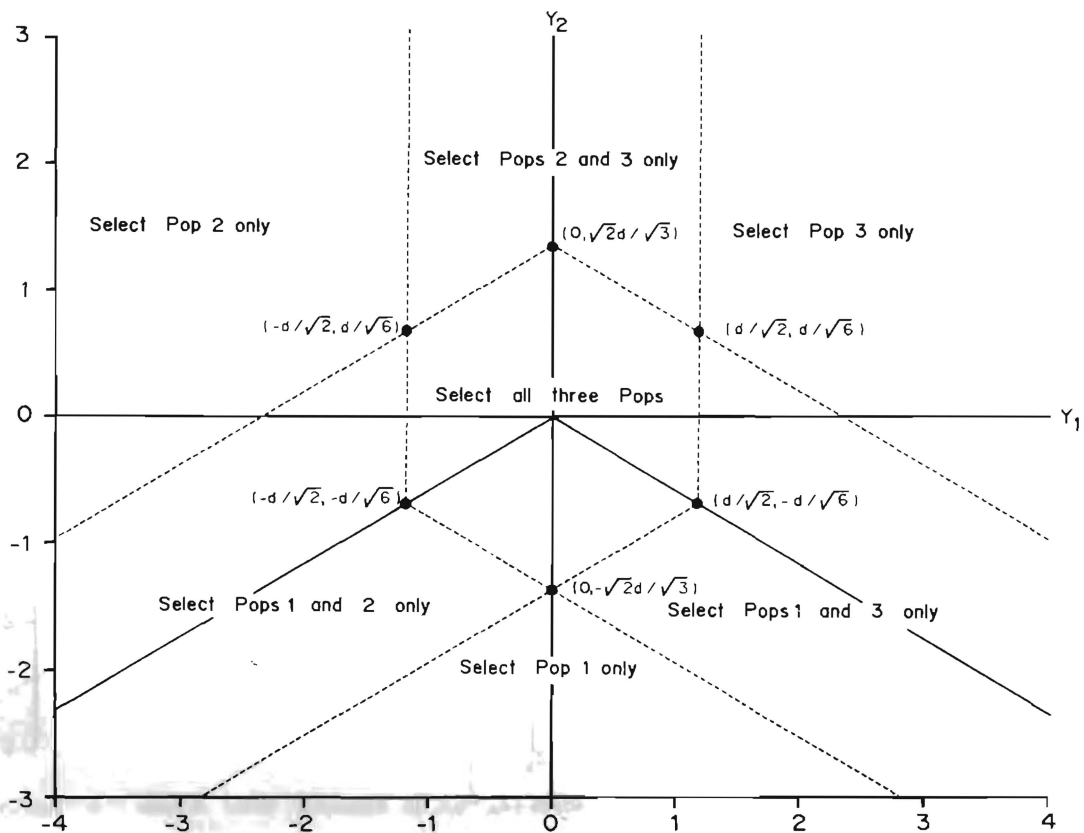


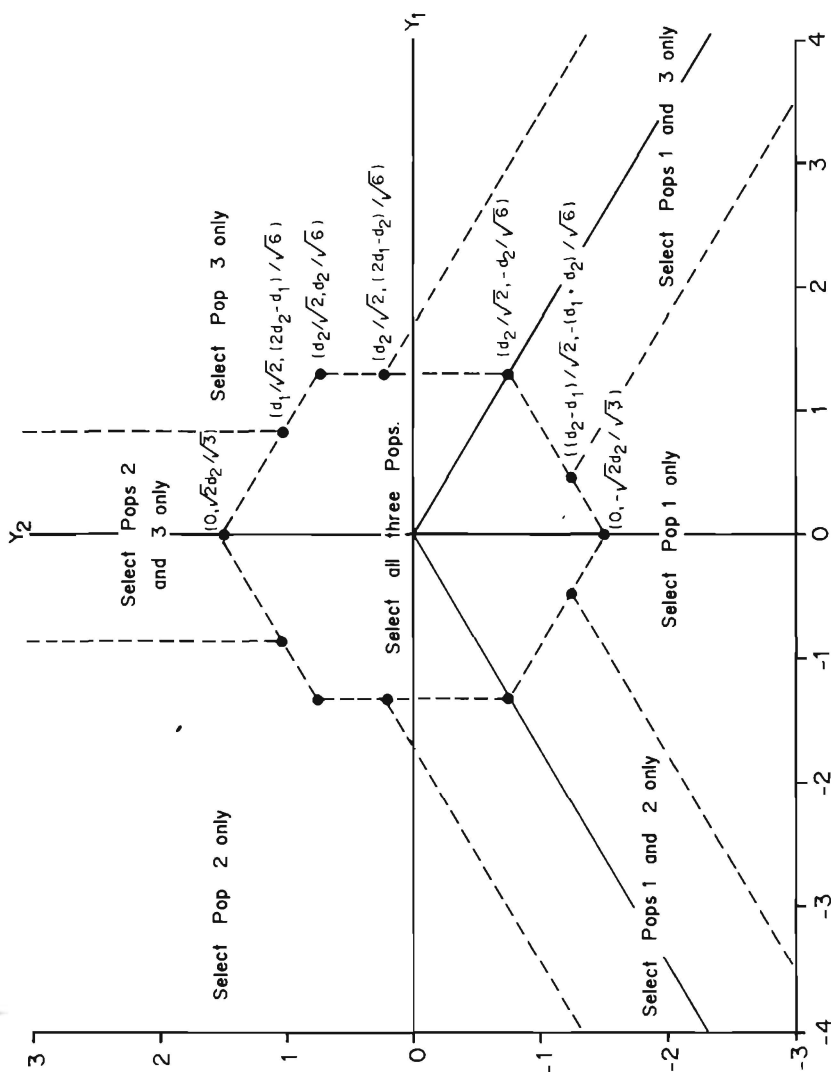
FIGURE 2. Multiple range procedure in (Y_1, Y_2) -space

FIGURE 3. Meaning of v-configurations

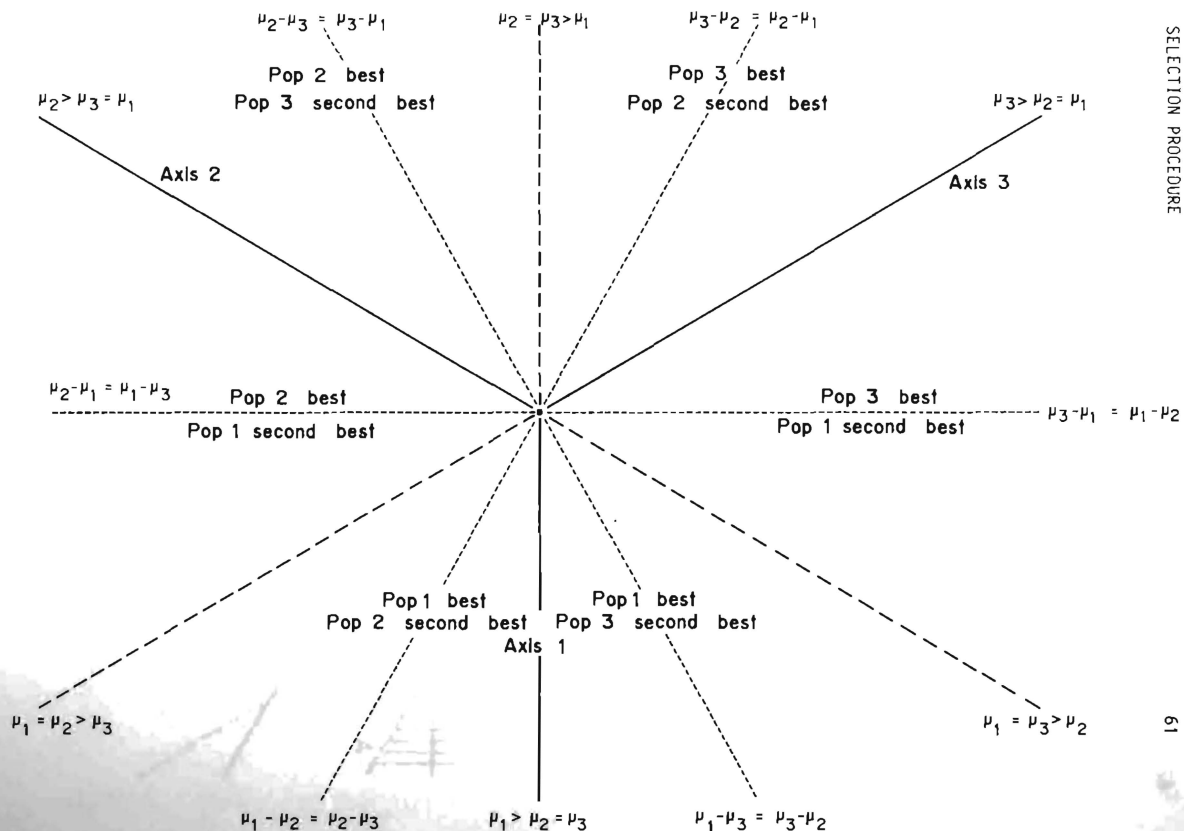


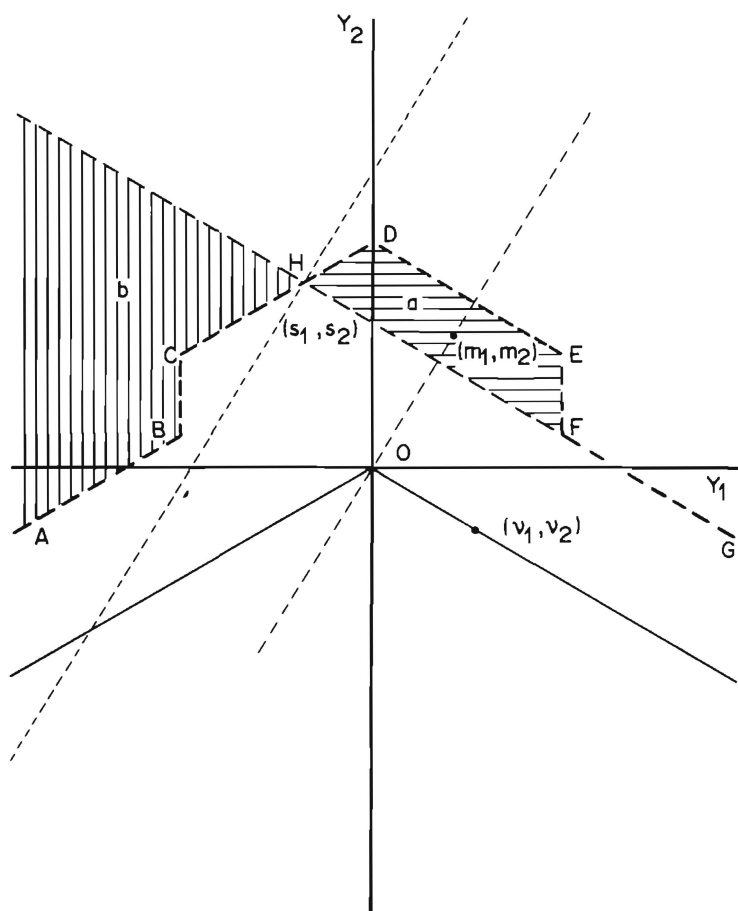
FIGURE 4. Proof that $PCS \geq P^*$ 

FIGURE 5. Comparison of PCS of Gupta and multiple range procedures

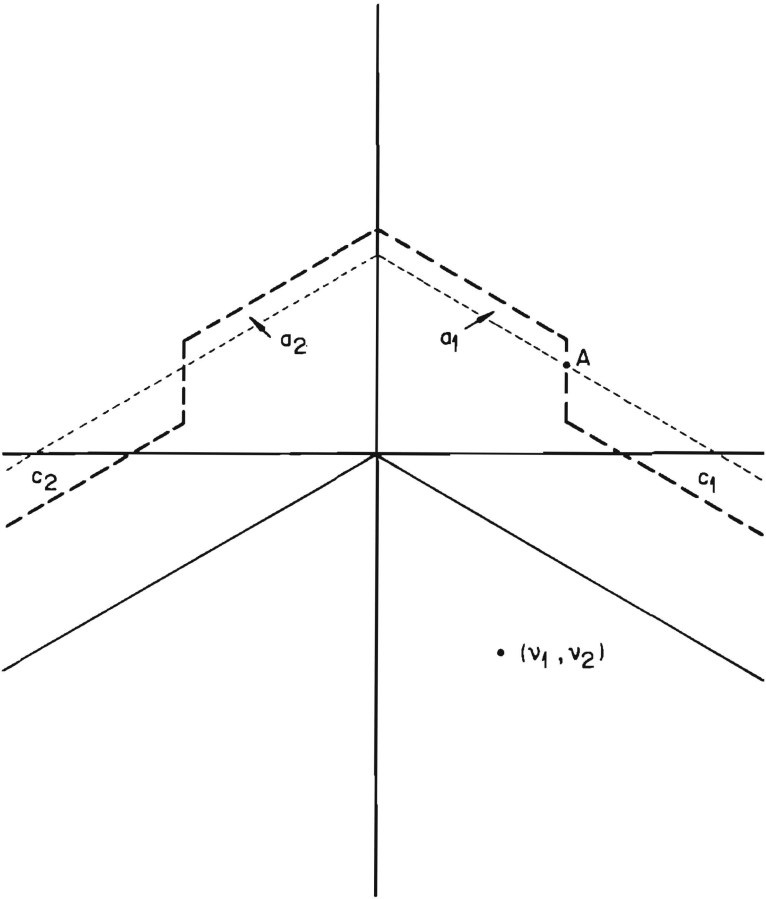


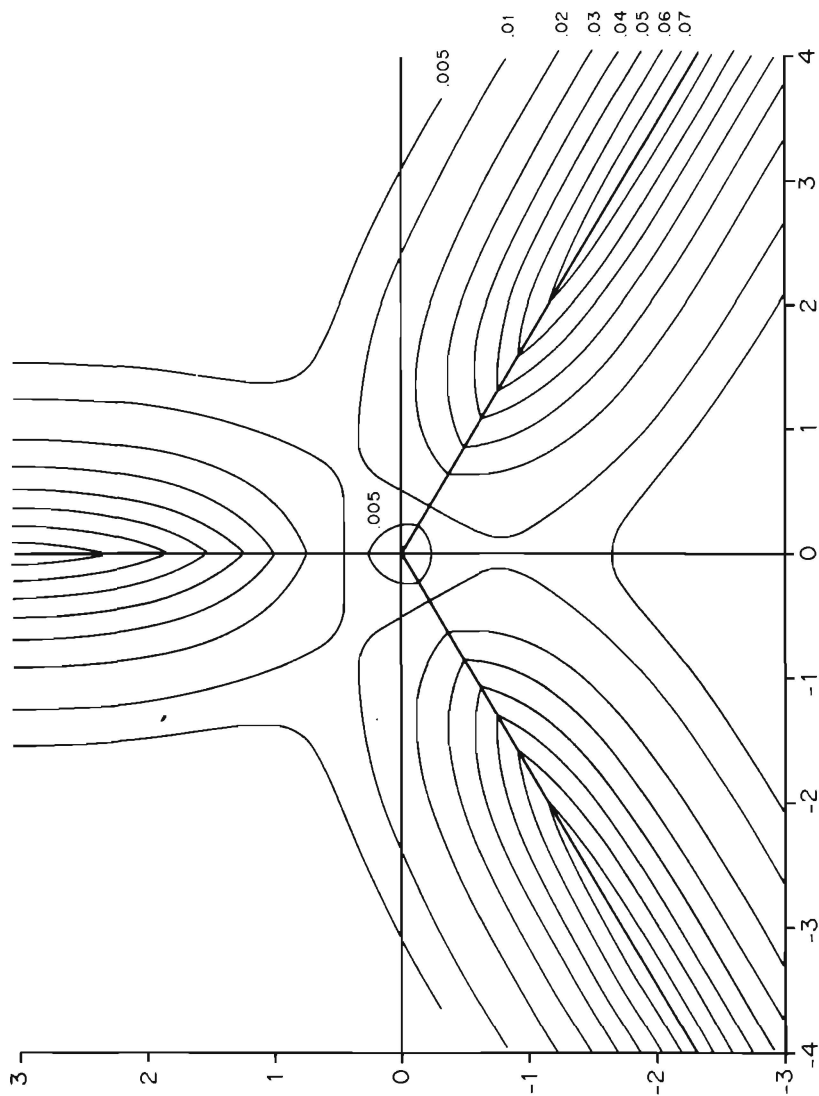
FIGURE 6. Contours of difference between PCS's at $P^* = 0.8$ 

FIGURE 7a. Derivation of expression for ESS of Gupta's procedure

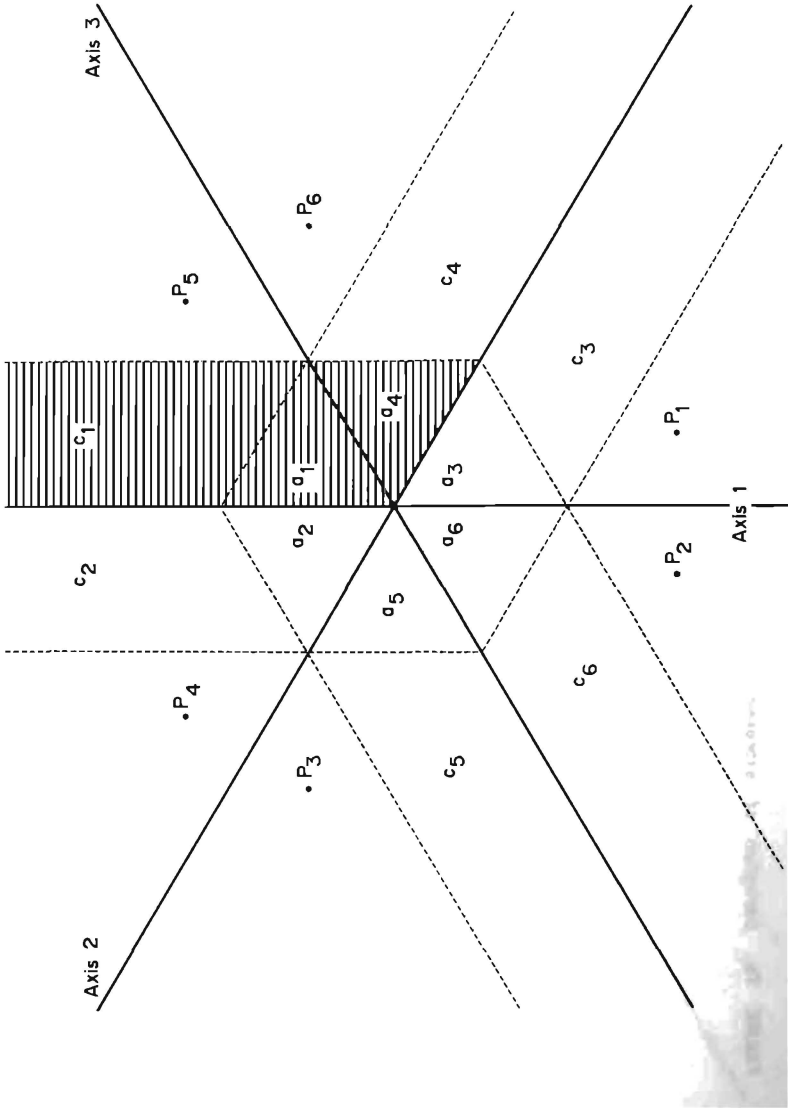


FIGURE 7b. Derivation of expression for ESS of multiple range procedure

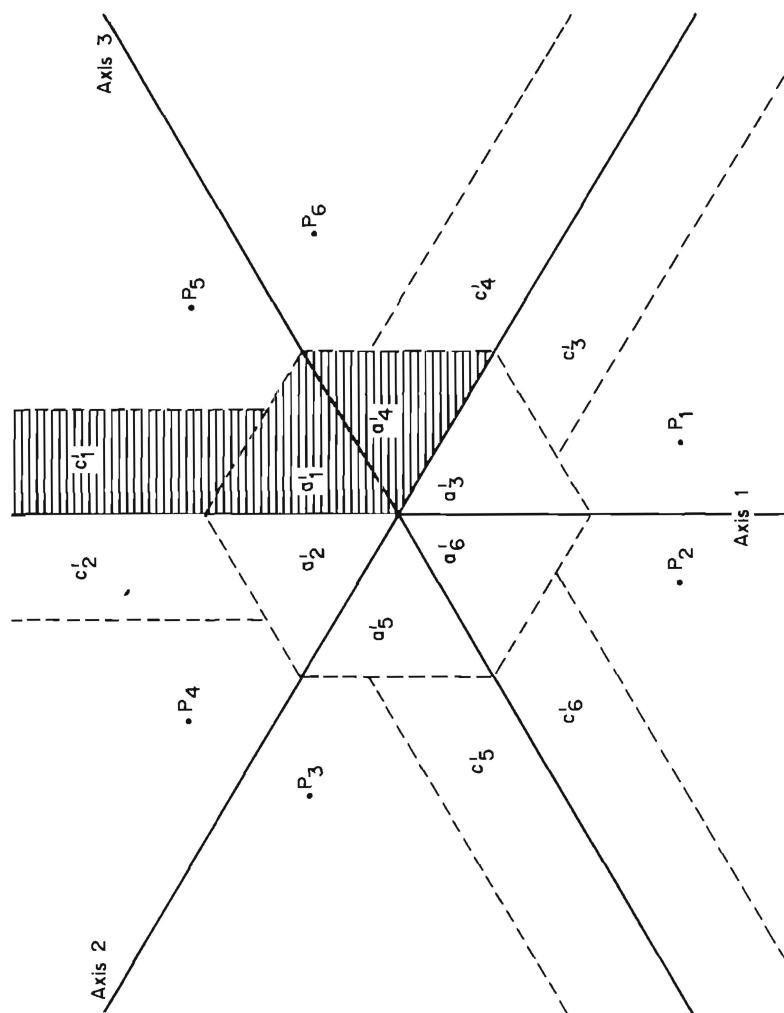


FIGURE 7c. Difference between ESS of Gupta's and multiple range procedures

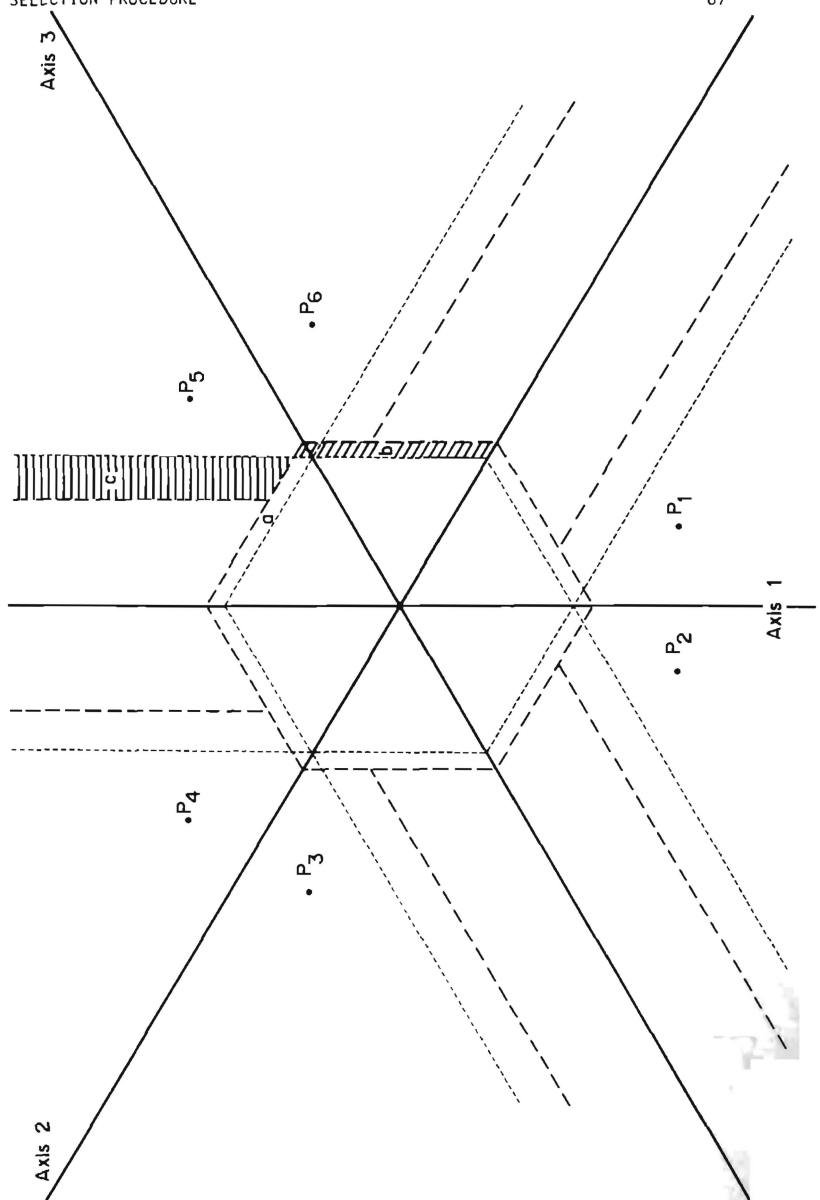


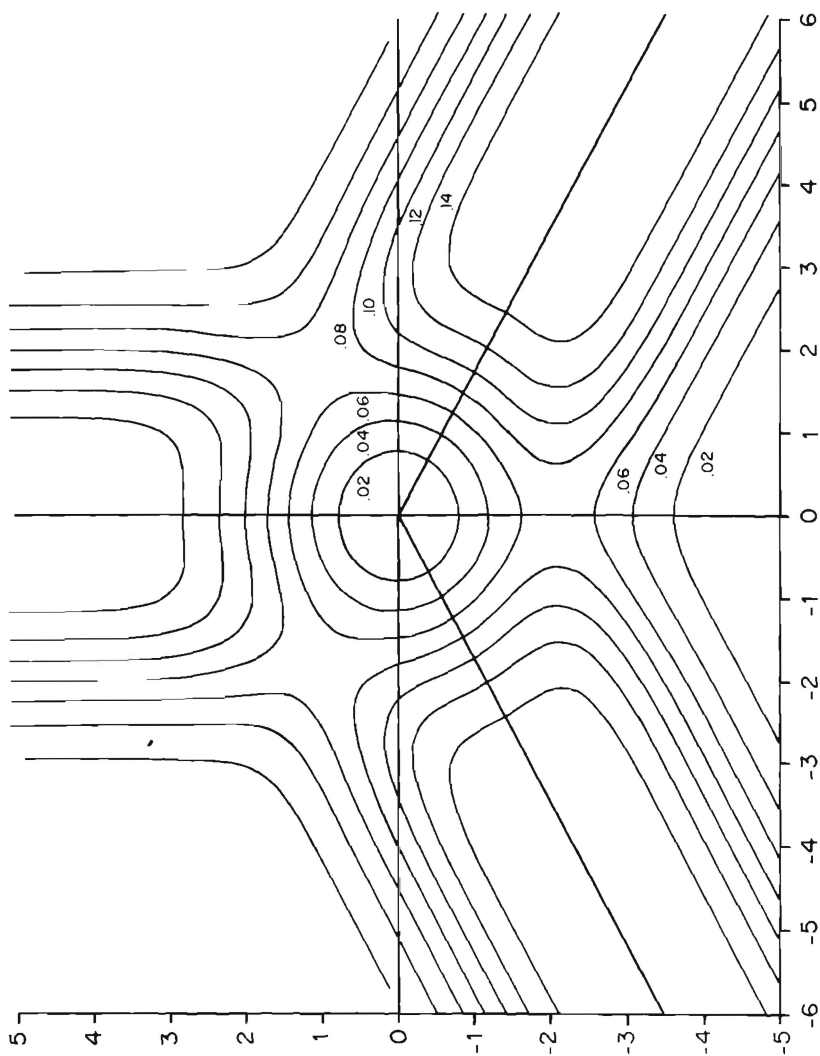
FIGURE 8. Contours of difference between ESS's at $P^* = 0.8$ 

FIGURE 9. Contours of difference between ESS's at $P^* = 0.99$

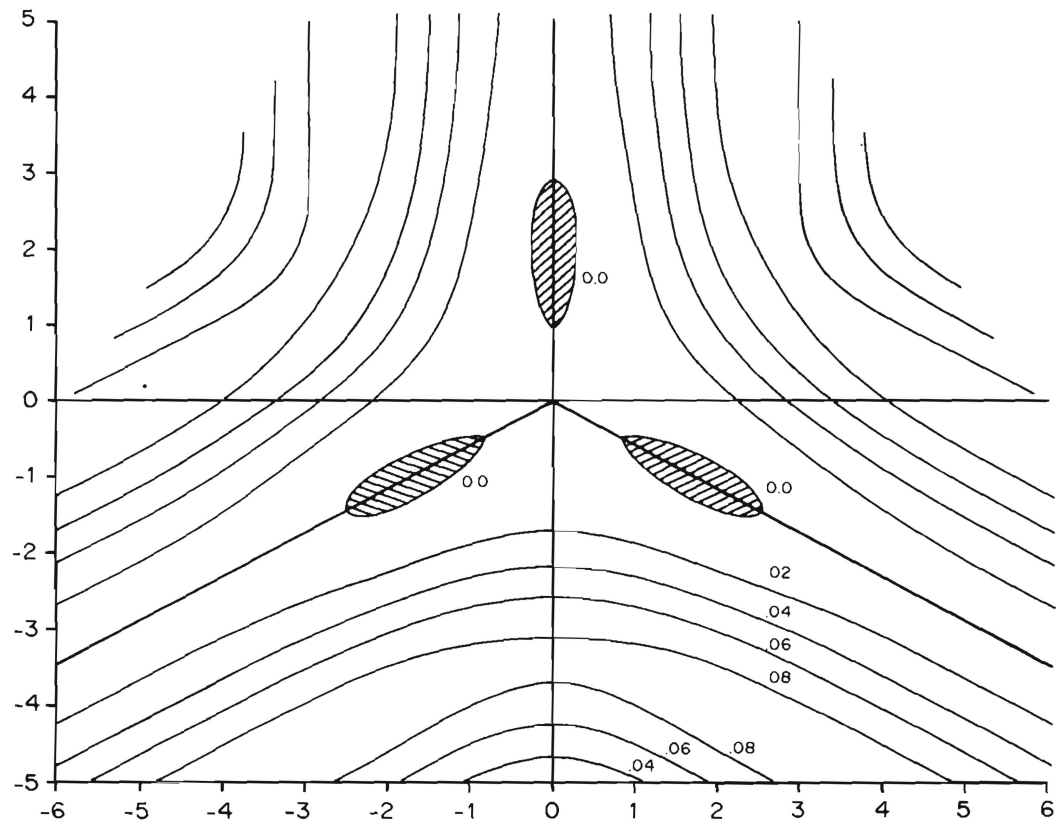


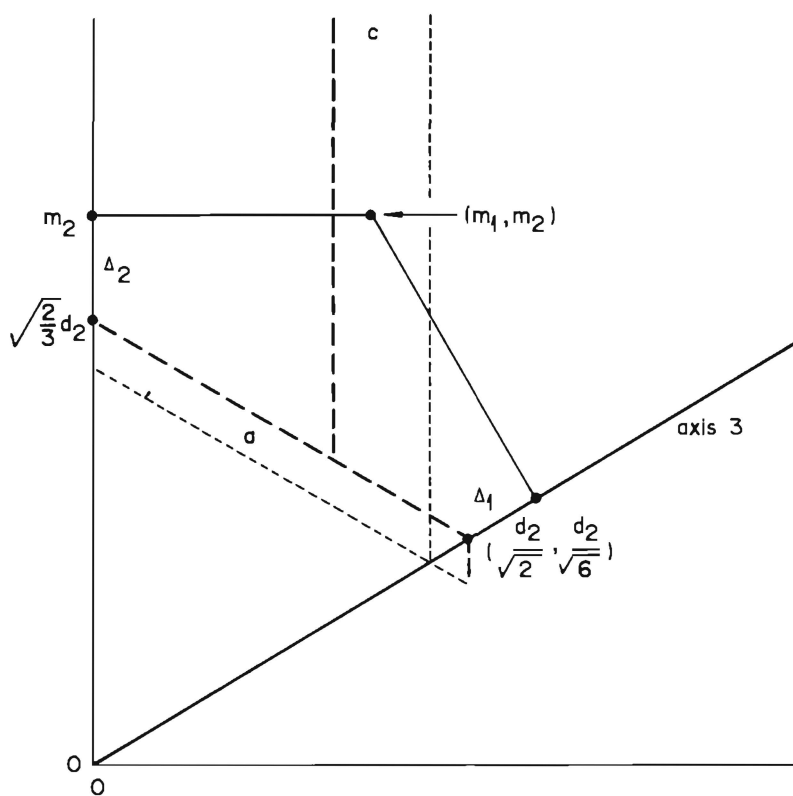
FIGURE 10. Sufficient Conditions for $D \geq 0$ 

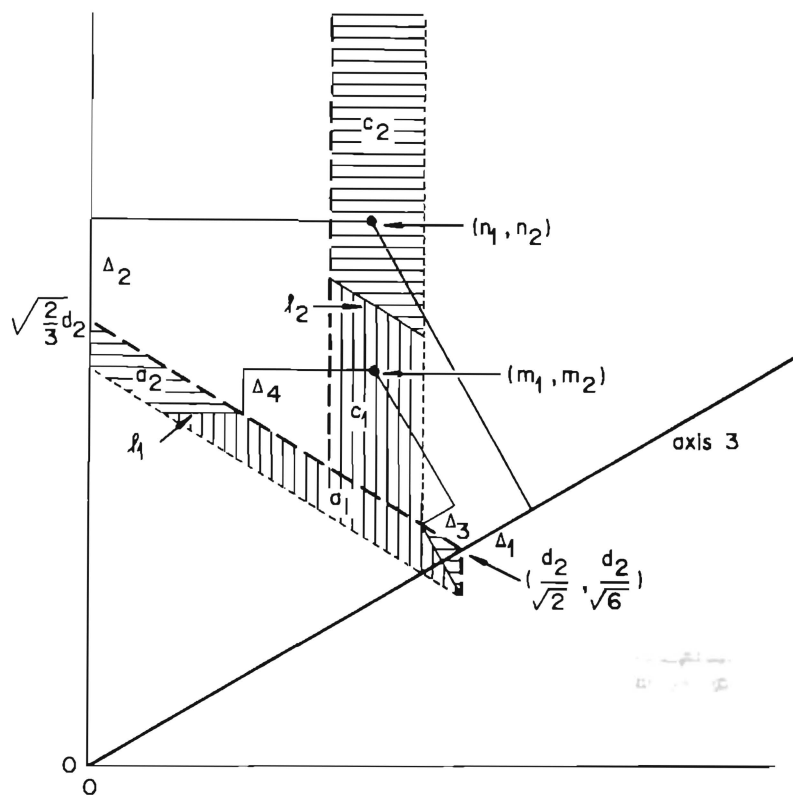
FIGURE 11. Sufficient Conditions for $D \geq 0$ 

FIGURE 12. Δ^* as a function P^* 